A. Noldbach problem

time limit per test: 2 seconds
memory limit per test: 64 megabytes

Nick is interested in prime numbers. Once he read about Goldbach problem. It states that every even integer greater than 2 can be expressed as the sum of two primes. That got Nick's attention and he decided to invent a problem of his own and call it Noldbach problem. Since Nick is interested only in prime numbers, Noldbach problem states that at least k prime numbers from 2 to n inclusively can be expressed as the sum of three integer numbers: two neighboring prime numbers and 1. For example, 19 = 7 + 11 + 1, or 13 = 5 + 7 + 1.

Two prime numbers are called neighboring if there are no other prime numbers between them.

You are to help Nick, and find out if he is right or wrong.

Input

The first line of the input contains two integers n ($2 \le n \le 1000$) and k ($0 \le k \le 1000$).

Output

Output YES if at least k prime numbers from 2 to n inclusively can be expressed as it was described above. Otherwise output NO.

Examples

input Copy

```
27 2
```

output Copy

```
YES
```

input Copy

```
45 7
```

output Copy

```
NO
```

Note

In the first sample the answer is YES since at least two numbers can be expressed as it was described (for example, 13 and 19). In the second sample the answer is NO since it is impossible to express 7 prime numbers from 2 to 45 in the desired form.

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 17
Finished
Practice

→ Virtual participation

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Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
370486663	Apr/10/2026 05:02	Accepted
370486426	Apr/10/2026 04:56	Accepted
370486385	Apr/10/2026 04:54	Compilation error

→ Problem tags

brute force | math | number theory | *1000

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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